MIT Assignment-10

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- Given are two numbera. Using 8086 instruction set

Q1:

Add, substract the given numbers CODE:

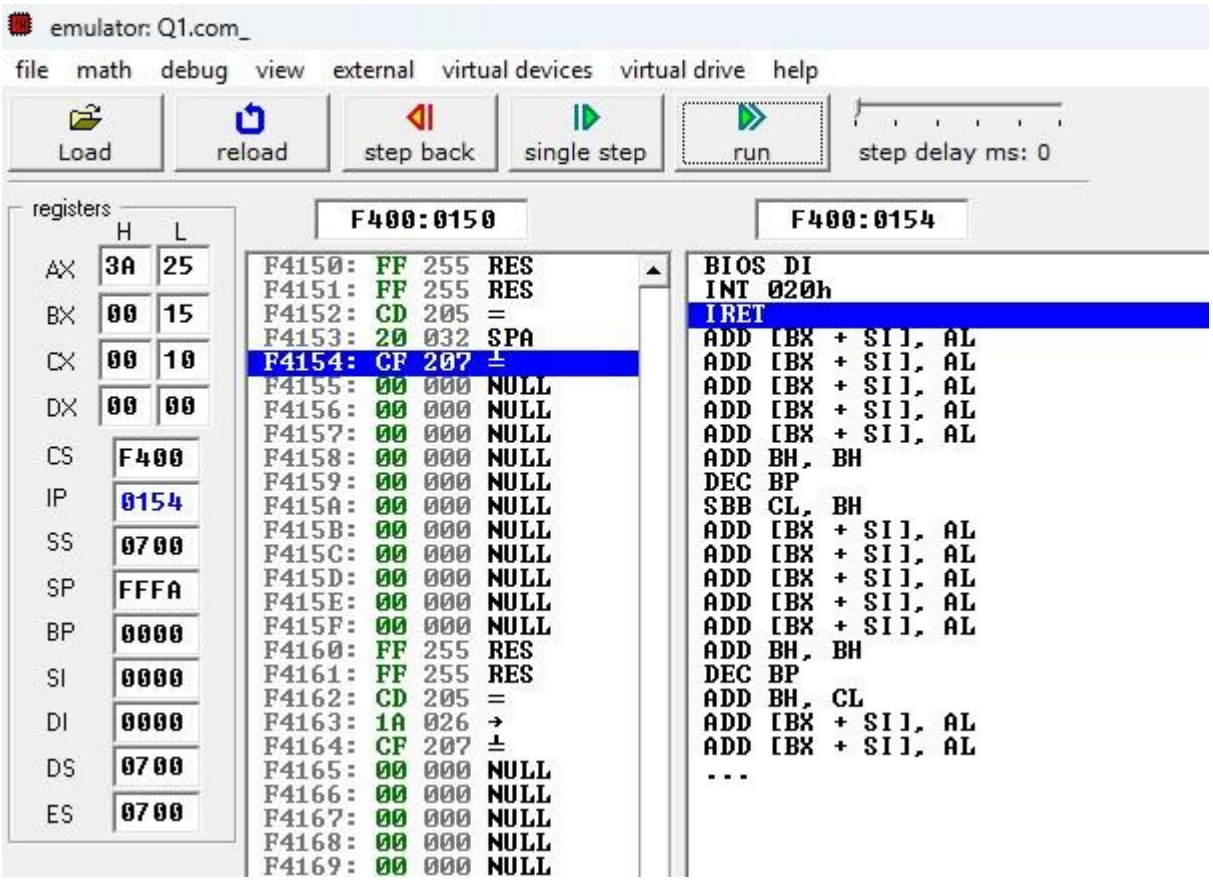
```
; You may customize this and other start-up templates;
; The location of this template is c:\emu8086\inc\0_com_template.txt

org 100h

MOV AL, 25H      ; 1st number
MOV BL, 15H      ; 2ND number

; ADD
MOV AH, AL
ADD AH, BL       ; AH = AL + BL

; SUB
```

MOV CL, AL

SUB CL, BL ; CL = AL - BL

ret

OUTPUT:



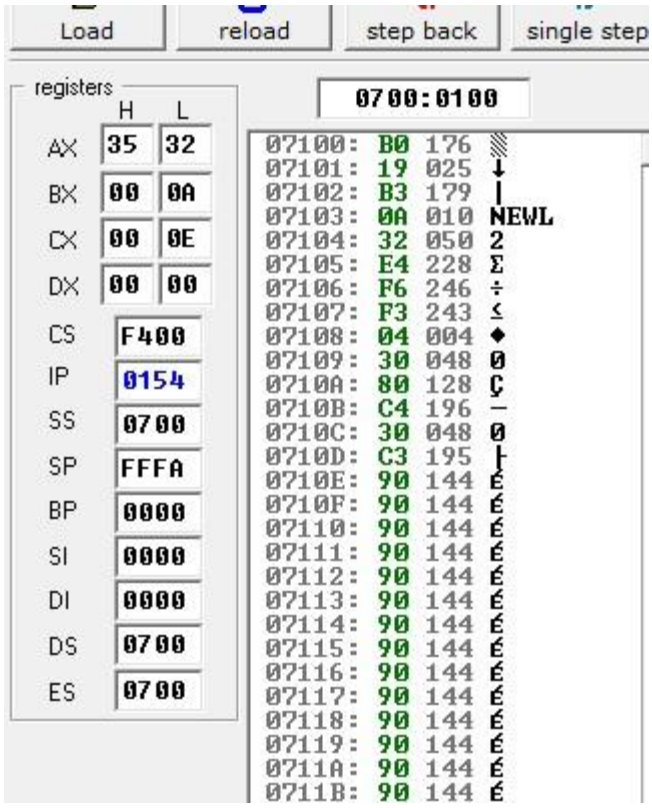
Q2:

Convert the numbers into ASCII

CODE:

```
org 100h

MOV AL, 25      ; Number to convert (decimal 25) -> ASCII
MOV BL, 10      ; Divisor

;  Convert number in AL to ASCII
XOR AH, AH      ; Clear AH for DIV
DIV BL          ; AL / 10 -> Quotient in AL, Remainder in AH
ADD AL, '0'     ; Convert quotient to ASCII, stored in AL
ADD AH, '0'     ; Convert remainder to ASCII, stored in AH

; Now:
; AL = ASCII of tens digit
; AH = ASCII of units digit

ret
```

OUTPUT:



Q3:

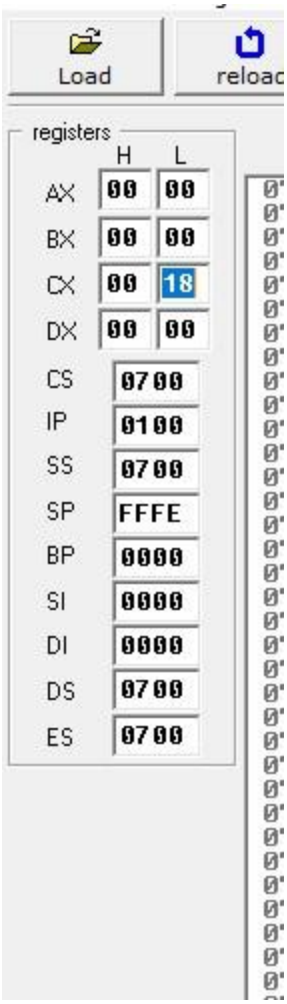
Add 47H to output of addition. If a carry is generated then display FFH else display 00H in register CX.

CODE:

```
org 100h


MOV AL, 25H      ; 1st number

MOV BL, 15H      ; 2nd number


; ADD

MOV AH, AL
```

```
ADD AH, BL       ; AH = AL + BL

; --- Add 47H and check carry ---
MOV AL, AH       ; Load addition result
ADD AL, 47H      ; Add 47H
JC  carry_flag   ; Jump if carry generated

; --- No carry ---
MOV CX, 0000H    ; Set CX = 0000H if no carry
JMP done

carry_flag:
MOV CX, 0FFFFH   ; Set CX = FFFFH (FFH) if carry

done:
ret

ret
```

OUTPUT:


Load
reload
registers
H
L
AX 00 00
BX 00 00
CX 00 18
DX 00 00
CS 0700
IP 0100
SS 0700
SP FFFE
BP 0000
SI 0000
DI 0000
DS 0700
ES 0700